DATA ANALYSIS

LIFE EXPECTANCY

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CONTEXT OF **STUDY**

The Global Health Observatory (GHO) dataset is an examination of worldwide trends in health between 2000 and 2015. The dataset contains information about life expectancy and different determinants for 193 nations, specifically immunization coverage, mortality rates, economic metrics, and social determinants of health. The purpose of this project is to predict which specific factors significantly increase or decrease life expectancy, including alcohol consumption, infant mortality rates, country's status, GDP, adult mortality rates, etc.

HYPOTHESIS

The general hypothesis is to prove that the variables just mentioned have a significant impact on life expectancy.

- H0= There is no relationship between the dependent variable (life expectancy) and the independent variables

- Ha= There is a relationship between the dependent variable (life expectancy) and the independent variables

THE GOALS OF THE STUDY

- Identifying Key Influences: Describing how immunization, mortality, economic status, and social influences contribute to overall health outcomes.

- Differences between Developed and Developing Countries: Observing the distinct impact of developing countries on health progress by analyzing the main differences in social and economic factors.

- Explaining Economic and Social Effects: Describing how health investments play a role in influencing trends in public health.

STUDY
DESIGN

My dependant variable

My dependent variable is life expectancy since it is a continuous measurable variable.

Problems I might encounter

- Multicollinearity
- Influential Outliers
- No correlation between variables
- Other external influences not included in the datset

POPULATION AND **ELEMENTS**

Population from which the subjects were selected:

Since I did not collect this data, the explanation of where and how the observational study was conducted is abstract.

Describe the sampling procedure:

The sampling procedure they seemed to use was clustering. If you look in the dataset, there is the categorical variable called country that ranges from Afghanistan to Zimbabwe. They clustered the customers by country and then conducted a simple random sampling in each country to obtain their sample population. Given the amount of data and limitation of my PC to process this many information. I had to exclude a couple of columns to be able to proceed with my investigation. However, I will be using the entire number of rows for this project

LINEAR MODEL

The linear model for the full model (with all variables) is

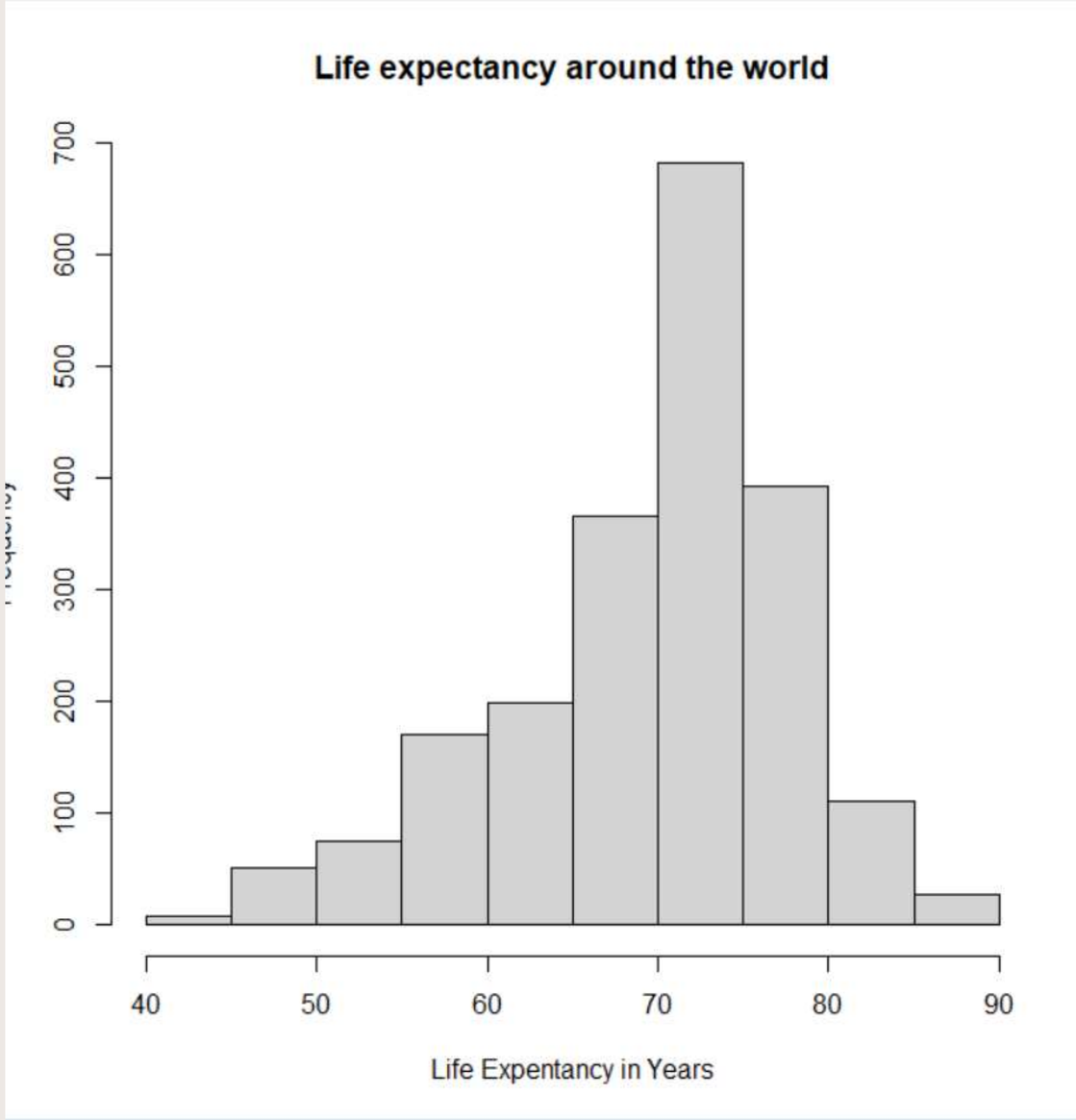
$$y_i = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \beta_4 x_4 + \beta_5 x_5 + \beta_6 x_6 + \beta_7 x_7 + \beta_8 x_8 + \beta_9 x_9 + \beta_{10} x_{10} + \beta_{11} x_{11} + \beta_{12} x_{12} + \beta_{13} x_{13} + \beta_{14} x_{14} + \beta_{15} x_{15} + \varepsilon_i$$

Where y_i: response (Life Expectancy), β_0: intercept, β_1 - β_{15}: partial slopes (coefficients),

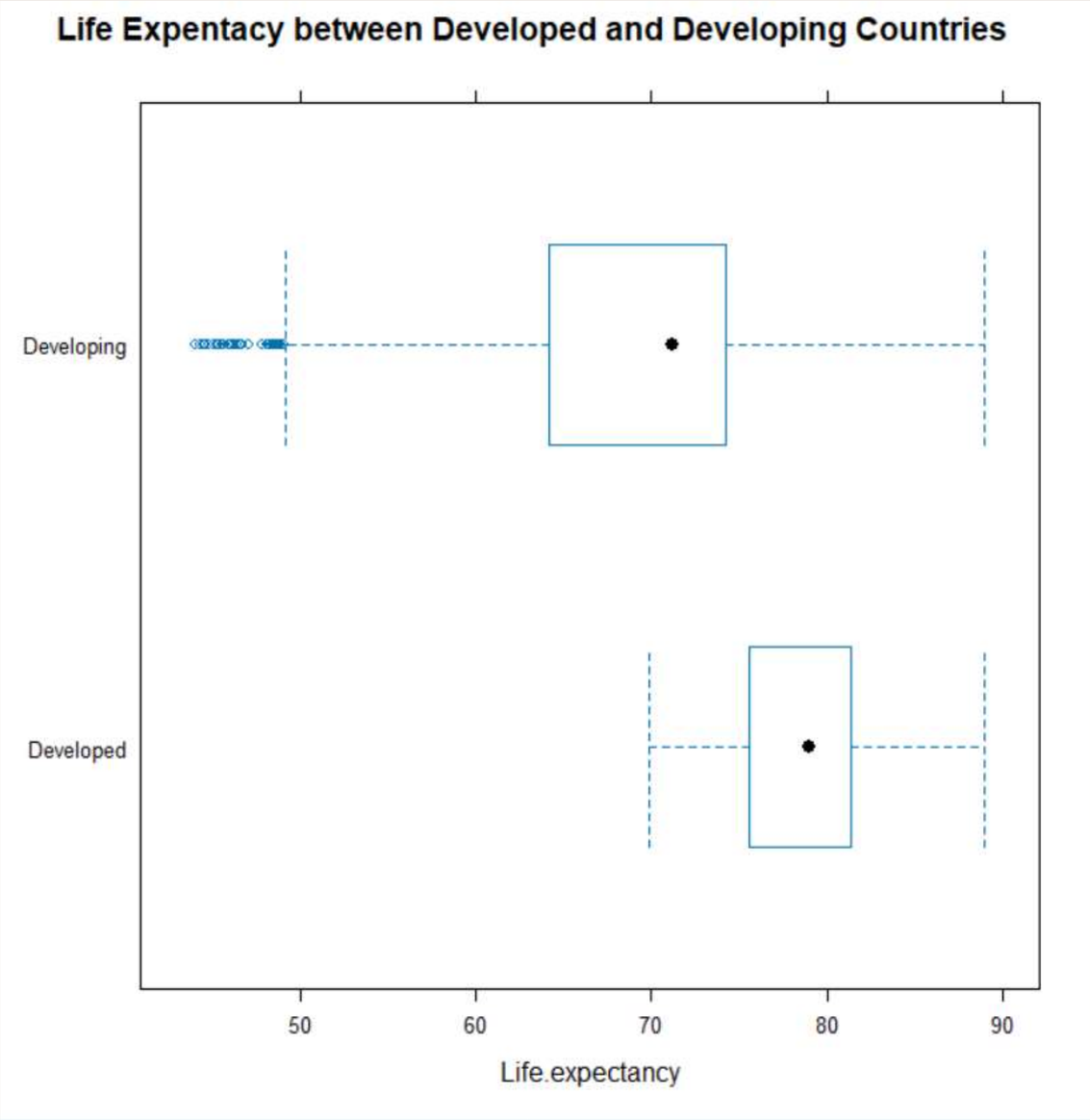
x1:Status , x2: Adult.Mortality , x3:infant.deaths, x4: Alcohol, x5: % expenditure, x6:Hepatitis.B ,x7:Measles , x8:BMI, x9:under.five.deaths, x10: Polio, x11:Total.expenditure, x12:Diphtheria ,x13: HIV.AIDS,x14:Income.composition.of.resources, x15:Schooling and ε_i: residual (error) term

GRAPHS

The dependant variable is normally distributed

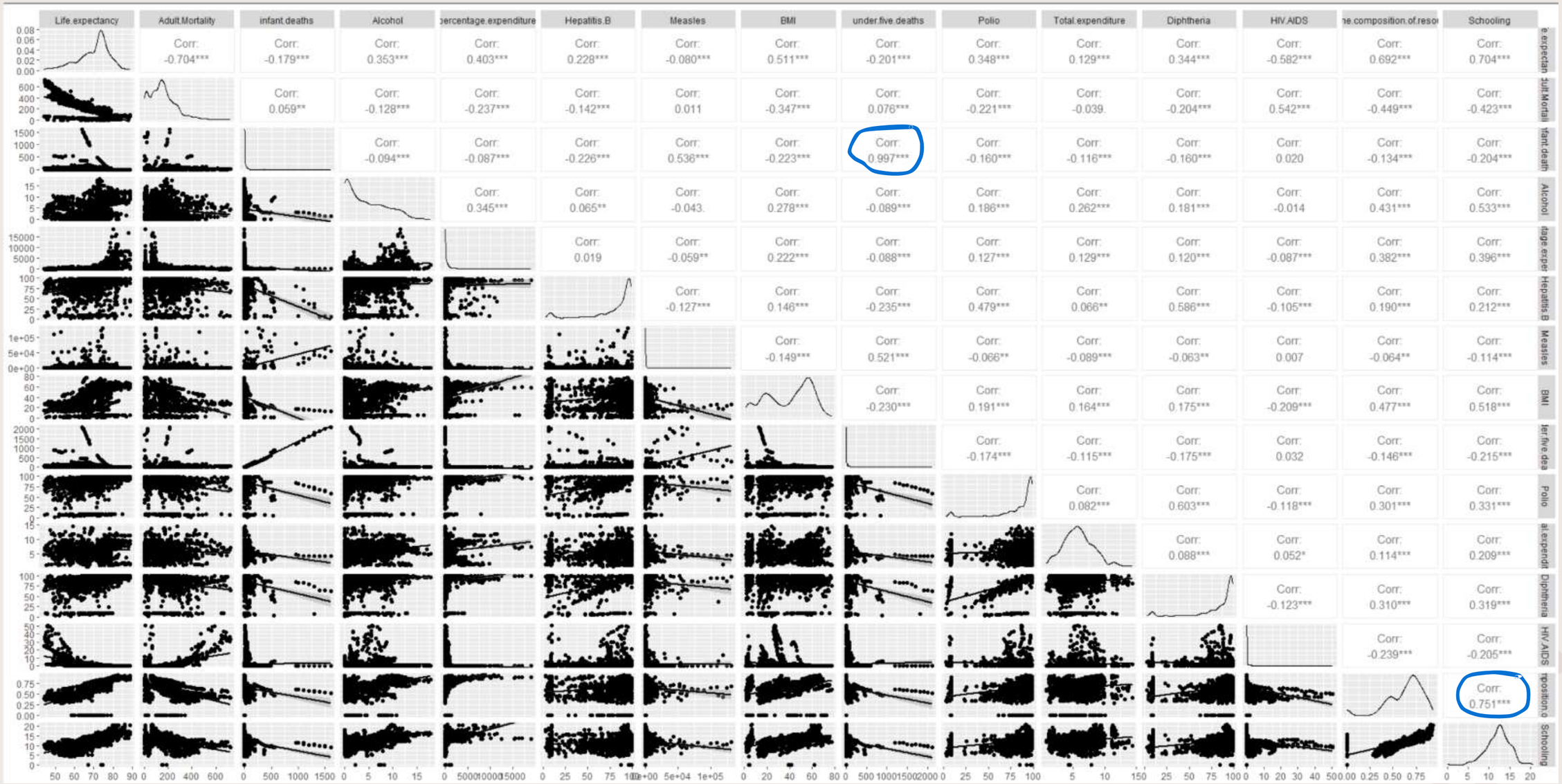


There seems to be a significant difference in life expctancy between developed countries and developing countries



GRAPHS

I circled the variables that are highly correlated between each other that can cause multicollinearity issues



FIRST MODEL

fit0=lm(Life.expectancy~.,data=lmlife.data)
summary(fit0)

```
Residuals:
     Min       1Q   Median       3Q      Max 
-18.2962  -2.0185  -0.0375   2.1904  18.6925 

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)    
(Intercept)                     5.602e+01  7.173e-01  78.100  < 2e-16 ***
StatusDeveloping               -9.676e-01  3.002e-01  -3.223  0.00129 ** 
Adult.Mortality                -1.806e-02  8.951e-04 -20.175  < 2e-16 ***
infant.deaths                   9.930e-02  1.012e-02   9.808  < 2e-16 ***
Alcohol                         3.779e-02  2.683e-02   1.409  0.15909    
percentage.expenditure          4.622e-04  5.472e-05   8.446  < 2e-16 ***
Hepatitis.B                    -3.030e-03  4.059e-03  -0.747  0.45543    
Measles                        -8.066e-06  1.060e-05  -0.761  0.44676    
BMI                             3.253e-02  4.881e-03   6.666 3.37e-11 ***
under.five.deaths              -7.594e-02  7.460e-03 -10.180  < 2e-16 ***
Polio                           1.465e-02  4.796e-03   3.054  0.00228 ** 
Total.expenditure               5.784e-02  3.588e-02   1.612  0.10718    
Diphtheria                      1.630e-02  5.345e-03   3.050  0.00232 ** 
HIV.AIDS                       -4.544e-01  1.746e-02 -26.016  < 2e-16 ***
Income.composition.of.resources 8.090e+00  6.849e-01  11.813  < 2e-16 ***
Schooling                       7.231e-01  4.914e-02  14.715  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.6 on 2067 degrees of freedom
Multiple R-squared:  0.8137,	Adjusted R-squared:  0.8123 
F-statistic: 601.8 on 15 and 2067 DF,  p-value: < 2.2e-16
```

By focusing on the p-values it is noticeable which independent variables are not significant enough to what we would like to prove. This variables are Alcohol, Hepatitis B, Measles and Total Expenditure which all resulted in p-values>0.05

OVERALL
ANALYSIS OF MODEL

H0: $\beta 1 = \beta 2 = \beta 3 = \beta 4 \ldots \beta 15$

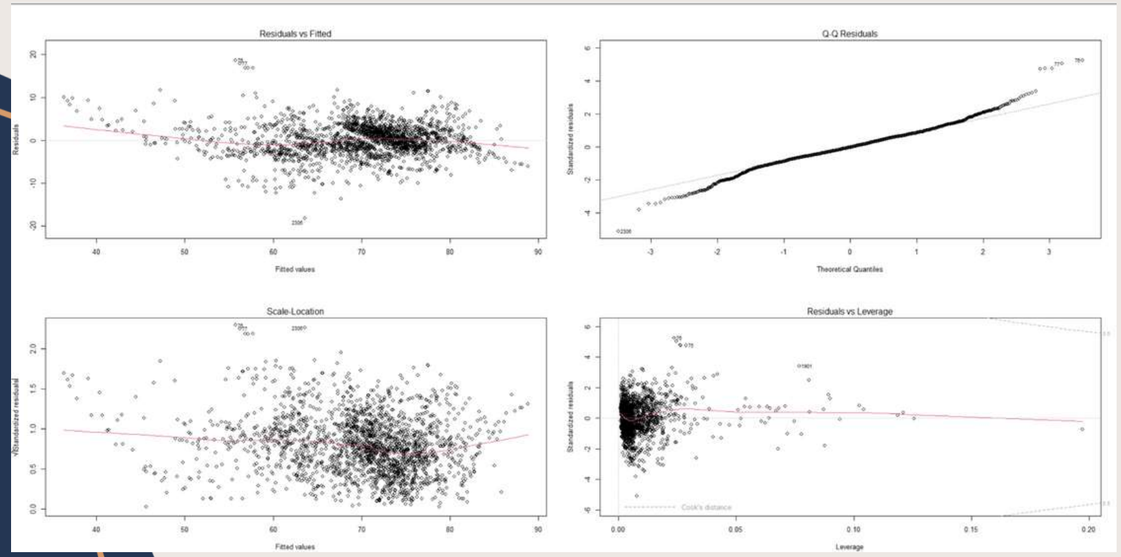
Ha : H0 is not true, (or at least one $\beta j = 0$)

F =601.8, df1 = 15, df2 =2067 , pvalue = 2.2e-16 ≈ 0. Since the pvalue ≤ α (0.05), the null is rejected so

the conclusion is that the predictors collectively contribute to explaining the variability in the dependent

variable

The R2 adj = 0.81 ≥ 0.6 , which is a good sign



ALL ASSUMPTIONS ARE MET!

PROBLEMS WITH THE MODEL

```
vif(fit0) ##Infant deaths and under.five.deaths have high correlation
          Status                  Adult.Mortality
                1.721240                         1.795614
             infant.deaths                          Alcohol
                192.741748                         1.834356
percentage.expenditure                       Hepatitis.B
                1.353571                         1.651147
                 Measles                              BMI
                1.473040                         1.482494
       under.five.deaths                            Polio
                190.129950                         1.704641
       Total.expenditure                       Diphtheria
                1.113729                         2.004829
                 HIV.AIDS Income.composition.of.resources
                1.439446                         2.562997
                Schooling
                 3.037288

res0=residuals(fit0)
dwt(res0)
0.7103734
```

There are highly correlated variables within the model, which can make it difficult to determine the individual impact of each variable. Additionally, this can introduce bias, affecting the accuracy of the results.

DWT= 0,71 is not inside the 1<DWT<3 indicating that residuals are dependent.

MODEL SELECTION

step(fit0
)

fit1=lm(Life.expectancy ~ Status + Adult.Mortality + infant.deaths + Alcohol + percentage.expenditure + BMI + under.five.deaths + Polio + Total.expenditure + Diphtheria + HIV.AIDS + Income.composition.of.resources + Schooling, data=lmlife.data)
summary(fit1)

```
Residuals:
     Min       1Q   Median       3Q      Max 
-18.2735  -2.0432  -0.0555   2.1897  18.6867 

Coefficients:
                                  Estimate Std. Error t value Pr(>|t|)    
(Intercept)                     55.9375851  0.7120236  78.561  < 2e-16 ***
StatusDeveloping                -0.9579042  0.2995592  -3.198  0.00141 ** 
Adult.Mortality                 -0.0180554  0.0008948 -20.178  < 2e-16 ***
infant.deaths                    0.0976521  0.0098030   9.961  < 2e-16 ***
Alcohol                          0.0382862  0.0266802   1.435  0.15144    
percentage.expenditure           0.0004661  0.0000545   8.553  < 2e-16 ***
BMI                              0.0326303  0.0048733   6.696 2.76e-11 ***
under.five.deaths               -0.0749153  0.0072996 -10.263  < 2e-16 ***
Polio                            0.0139809  0.0047098   2.968  0.00303 ** 
Total.expenditure                0.0582567  0.0358588   1.625  0.10440    
Diphtheria                       0.0146581  0.0048529   3.020  0.00255 ** 
HIV.AIDS                        -0.4542068  0.0174504 -26.028  < 2e-16 ***
Income.composition.of.resources  8.0908084  0.6846525  11.817  < 2e-16 ***
Schooling                        0.7233529  0.0491061  14.730  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.599 on 2069 degrees of freedom
Multiple R-squared:  0.8136,	Adjusted R-squared:  0.8124 
F-statistic: 694.6 on 13 and 2069 DF,  p-value: < 2.2e-16
```

The model that gave me the best AIC was the one on the left. However, after running the VIF test again I was still having multicollinearity issues. For this specific reason, even though this was the supposedly best model. I decided to run a couple more of models until I got rid of multicollinearity to have more predictable results.

FINAL MODEL

fit2=lm(Life.expectancy ~ Status + Adult.Mortality + infant.deaths +
Alcohol + percentage.expenditure + BMI +
Polio + Total.expenditure + Diphtheria + HIV.AIDS + Income.composition.of.resources +
Schooling, data=lmlife.data)
summary(fit2)

```
Residuals:
     Min       1Q   Median       3Q      Max
-18.4748  -2.0555  -0.0239   2.2104  19.2064

Coefficients:
                                 Estimate Std. Error t value Pr(>|t|)
(Intercept)                     5.502e+01  7.239e-01  75.997  < 2e-16 ***
StatusDeveloping               -9.252e-01  3.070e-01  -3.014 0.002613 **
Adult.Mortality                -1.901e-02  9.122e-04 -20.838  < 2e-16 ***
infant.deaths                  -2.651e-03  7.810e-04  -3.394 0.000702 ***
Alcohol                         1.700e-03  2.710e-02   0.063 0.949982
percentage.expenditure          4.492e-04  5.583e-05   8.045 1.44e-15 ***
BMI                             3.305e-02  4.994e-03   6.617 4.65e-11 ***
Polio                           1.661e-02  4.820e-03   3.446 0.000581 ***
Total.expenditure               5.786e-02  3.675e-02   1.574 0.115543
Diphtheria                      2.005e-02  4.945e-03   4.054 5.22e-05 ***
HIV.AIDS                       -4.594e-01  1.788e-02 -25.696  < 2e-16 ***
Income.composition.of.resources 8.385e+00  7.011e-01  11.961  < 2e-16 ***
Schooling                       7.486e-01  5.026e-02  14.894  < 2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 3.689 on 2070 degrees of freedom
Multiple R-squared:  0.8041,    Adjusted R-squared:  0.8029
F-statistic:   708 on 12 and 2070 DF,  p-value: < 2.2e-16
```

H0: $\beta 1 = \beta 2 = \beta 3 = \beta 4 \ldots \beta 15$

Ha : H0 is not true, (or at least one $\beta j = 0$)

F =708, df1 = 12, df2 =2070 , pvalue = 2.2e-16 ≈ 0. Since the pvalue ≤ α (0.05), the null is rejected so the conclusion is that the predictors collectively contribute to explaining the variability in the dependent variable

The R2 adj = 0.8029 ≥ 0.6 , which is a good sign

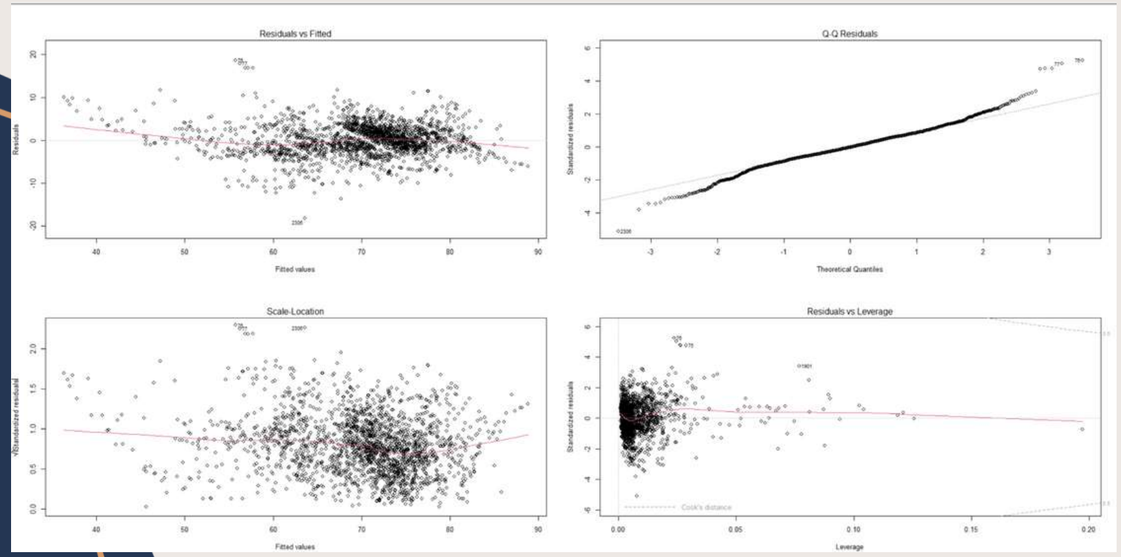
OVERALL
ANALYSIS OF FINAL MODEL

vif(fit2)

```
> vif(fit2)
                        Status                   Adult.Mortality
                      1.713900                          1.775802
                  infant.deaths                           Alcohol
                      1.092430                          1.782938
         percentage.expenditure                               BMI
                      1.341909                          1.478453
                         Polio                 Total.expenditure
                      1.639961                          1.112583
                    Diphtheria                          HIV.AIDS
                      1.634179                          1.436608
Income.composition.of.resources                        Schooling
                      2.557989                          3.026998
```

There is no multicollinearity left in the model. Even though the Adjusted R-squared decreased, the difference is not significant, and this model can eliminate biased results caused by the relationship between two independent variables.

```
> res2=residuals(fit2)
> dwt(res2)
[1] 0.7118557
```



- Linearity: Residual vs Fitted shows a very horizontal line which means that there is a linear
- relationship.
- Homoscedasticity: scale-location also shows a very horizontal line with no concerning increases or
- decreases indicating variance of residuals.
- Independence: DWT= 0.7 is not inside the 1<DWT<3 indicating that residuals are not independent.
- Normality: The QQ plot shows most points on the line suggesting normality of residuals
- Outliers: The last plot shows some outliers like numbers 78, 75,1905, but none seem to be influential.

CONCLUSIONS

01 Developing countries have a lower life expectancy than developed countries mostly because they lack resources and education.

02 Even though some of the independent variables are not significant it is important to keep them because they are communicating that alcohol consumption do not really affect life expectancy considering all the other variables.

03 Vaccination methods to combat sickness like Polio and Diphtheria show to be of importance to prevent early deaths in different countries.

04 Investments in education and general health resources are important. Meaning that highly educated countries have more expertise and capability to treat those who are sick.

Comparison to another research

I found a very interesting research that talks about how life expectancy has been slowly starting to decrease in 20 of 23 OECD countries due to different factors. Even though this research is based on the aging of expectancy, some of their conclusions align very closely with mine regarding why developing countries tend to have lower life expectancy than developed countries. It discusses how growing economic inequality in certain countries like the USA is causing people to miss the opportunity to receive proper care for sickness and end up with early deaths. This aligns with my variables explaining how countries with lower numbers in schooling and percentage expenditure in health resources have much lower years of life than those that possess the money and politics to invest in this situation.

Citations

Crimmins EM. Recent trends and increasing differences in life expectancy present opportunities for multidisciplinary research on aging. Nat Aging. 2021 Jan;1(1):12-13. doi: 10.1038/s43587-020-00016-0. Epub 2021 Jan 14. PMID: 34355199; PMCID: PMC8336715.